“Scheduling”

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Learning Outcome

- ❖ At the end of this lecture the students should be able to explain scheduling, priority rules of scheduling
- ❖ At the end of this lecture the students should be able to describe types of scheduling
- ❖ At the end of this lecture the students should be able to calculate different priority rules of scheduling
- ❖ At the end of this lecture the students should be able to solve Johnson’s Rule

Scheduling

Scheduling means Establishing the timing of the use of equipment, facilities, and human activities in an organization

Type of Scheduling

Job-shop scheduling

A work location in which a number of general purpose work stations exist and are used to perform a variety of jobs

Flow-shop scheduling

Flow-shop scheduling is a special case of job-shop scheduling where there is strict order of all operations to be performed on all jobs

Sequencing Jobs

- ☑ Specifies the order in which jobs should be performed at work centers
- ☑ Priority rules are used to dispatch or sequence jobs
  - ☑ FCFS: First come, first served
  - ☑ SPT: Shortest processing time
  - ☑ EDD: Earliest due date
  - ☑ LPT: Longest processing time

Priority Rules for Dispatching Jobs

First come, first served

The first job to arrive at a work center is processed first

Earliest due date

The job with the earliest due date is processed first

Shortest processing time

The job with the shortest processing time is processed first

Longest processing time

The job with the longest processing time is processed first

Sequencing Example

Apply the four popular sequencing rules to these five jobs

Job	*Job Work (Processing) Time (Days)*	*Job Due Date (Days)*
A	6	8
B	2	6
C	8	18
D	3	15
E	9	23

Sequencing Example

FCFS: Sequence A-B-C-D-E

Job Sequence	*Job Work (Processing) Time*	*Flow Time*	*Job Due Date*	*Job Lateness*
A	6	6	8	0
B	2	8	6	2
C	8	16	18	0
D	3	19	15	4
E	9	28	23	5
	28	77		11

Sequencing Example

FCFS: Sequence A-B-C-D-E

$$\textit{Average completion time} = \frac{\textit{Sum of total flow time}}{\textit{Number of jobs}} =$$

$$\textit{Utilization} = \frac{\textit{Total job work time}}{\textit{Sum of total flow time}} =$$

$$\textit{Average number of jobs in the system} = \frac{\textit{Sum of total flow time}}{\textit{Total job work time}} =$$

$$\textit{Average job lateness} = \frac{\textit{Total late days}}{\textit{Number of jobs}} =$$

Sequencing Example

FCFS: Sequence A-B-C-D-E

$$\textit{Average completion time} = \frac{\textit{Sum of total flow time}}{\textit{Number of jobs}} = 77/5 = 15.4 \textit{ days}$$

$$\textit{Utilization} = \frac{\textit{Total job work time}}{\textit{Sum of total flow time}} = 28/77 = 36.4\%$$

$$\textit{Average number of jobs in the system} = \frac{\textit{Sum of total flow time}}{\textit{Total job work time}} = 77/28 = 2.75 \textit{ jobs}$$

$$\textit{Average job lateness} = \frac{\textit{Total late days}}{\textit{Number of jobs}} = 11/5 = 2.2 \textit{ days}$$

Sequencing Example

SPT: Sequence B-D-A-C-E

Job Sequence	*Job Work (Processing) Time*	*Flow Time*	*Job Due Date*	*Job Lateness*
B	2	2	6	0
D	3	5	15	0
A	6	11	8	3
C	8	19	18	1
E	9	28	23	5
	28	65		9

Sequencing Example

SPT: Sequence B-D-A-C-E

$$\text{Average completion time} = \frac{\text{Sum of total flow time}}{\text{Number of jobs}} = 65/5 = 13 \textit{ days}$$

$$\text{Utilization} = \frac{\text{Total job work time}}{\text{Sum of total flow time}} = 28/65 = 43.1\%$$

$$\text{Average number of jobs in the system} = \frac{\text{Sum of total flow time}}{\text{Total job work time}} = 65/28 = 2.32 \textit{ jobs}$$

$$\text{Average job lateness} = \frac{\text{Total late days}}{\text{Number of jobs}} = 9/5 = 1.8 \textit{ days}$$

Sequencing Example

EDD: Sequence B-A-D-C-E

Job Sequence	Job Work (Processing) Time	Flow Time	Job Due Date	Job Lateness
B	2	2	6	0
A	6	8	8	0
D	3	11	15	0
C	8	19	18	1
E	9	28	23	5
	28	68		6

Sequencing Example

EDD: Sequence B-A-D-C-E

$$\textit{Average completion time} = \frac{\textit{Sum of total flow time}}{\textit{Number of jobs}} = 68/5 = 13.6 \textit{ days}$$

$$\textit{Utilization} = \frac{\textit{Total job work time}}{\textit{Sum of total flow time}} = 28/68 = 41.2\%$$

$$\textit{Average number of jobs in the system} = \frac{\textit{Sum of total flow time}}{\textit{Total job work time}} = 68/28 = 2.43 \textit{ jobs}$$

$$\textit{Average job lateness} = \frac{\textit{Total late days}}{\textit{Number of jobs}} = 6/5 = 1.2 \textit{ days}$$

Sequencing Example

LPT: Sequence E-C-A-D-B

Job Sequence	*Job Work (Processing) Time*	*Flow Time*	*Job Due Date*	*Job Lateness*
E	9	9	23	0
C	8	17	18	0
A	6	23	8	15
D	3	26	15	11
B	2	28	6	22
	28	103		48

Sequencing Example

LPT: Sequence E-C-A-D-B

$$\textit{Average completion time} = \frac{\textit{Sum of total flow time}}{\textit{Number of jobs}} = 103/5 = 20.6 \textit{ days}$$

$$\textit{Utilization} = \frac{\textit{Total job work time}}{\textit{Sum of total flow time}} = 28/103 = 27.2\%$$

$$\textit{Average number of jobs in the system} = \frac{\textit{Sum of total flow time}}{\textit{Total job work time}} = 103/28 = 3.68 \textit{ jobs}$$

$$\textit{Average job lateness} = \frac{\textit{Total late days}}{\textit{Number of jobs}} = 48/5 = 9.6 \textit{ days}$$

Sequencing Example

Summary of Rules

Rule	*Average Completion Time (Days)*	*Utilization (%)*	*Average Number of Jobs in System*	*Average Lateness (Days)*
FCFS	15.4	36.4	2.75	2.2
SPT	13.0	43.1	2.32	1.8
EDD	13.6	41.2	2.43	1.2
LPT	20.6	27.2	3.68	9.6

Comparison of Sequencing Rules

- ☑ No one sequencing rule excels on all criteria
- ☑ SPT does well on minimizing flow time and number of jobs in the system
- ☑ But SPT moves long jobs to the end which may result in dissatisfied customers
- ☑ FCFS does not do especially well (or poorly) on any criteria but is perceived as fair by customers
- ☑ EDD minimizes lateness

Sequencing N Jobs on Two Machines: Johnson's Rule

- ☑ Works with two or more jobs that pass through the same two machines or work centers
- ☑ Minimizes total production time and idle time

Johnson's Rule

1. List all jobs and times for each work center
2. Choose the job with the shortest activity time. If that time is in the first work center, schedule the job first. If it is in the second work center, schedule the job last.
3. Once a job is scheduled, it is eliminated from the list
4. Repeat steps 2 and 3 working toward the center of the sequence

Johnson's Rule Example

Job	*Work Center 1 (Drill Press)*	*Work Center 2 (Lathe)*
A	5	2
B	3	6
C	8	4
D	10	7
E	7	12

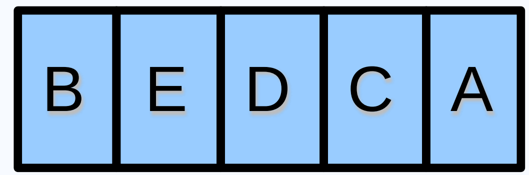
Johnson’s Rule Example

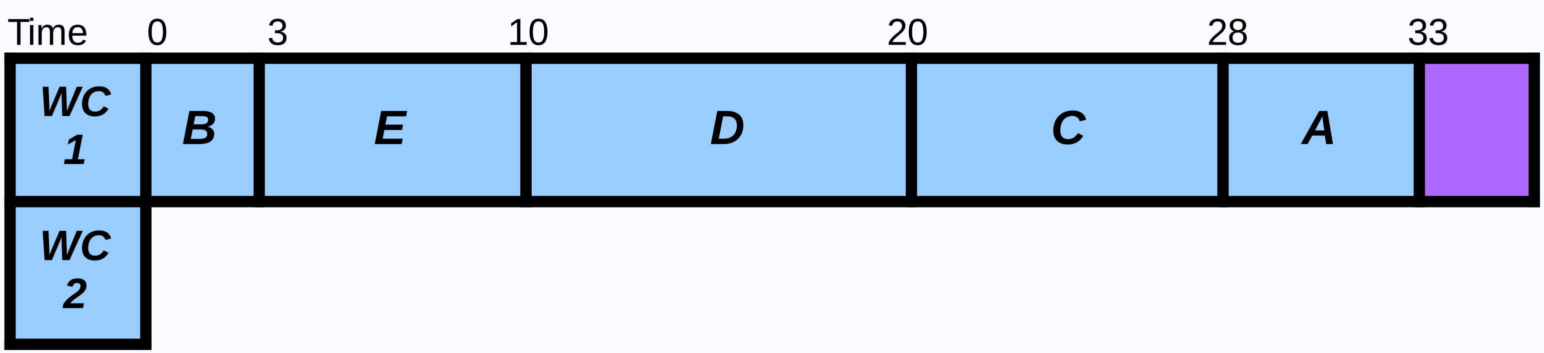
Job	*Work Center 1 (Drill Press)*	*Work Center 2 (Lathe)*
~~*A*~~	~~5~~	~~2~~
~~*B*~~	~~3~~	~~6~~
~~*C*~~	~~8~~	~~4~~
D	10	7
~~*E*~~	~~7~~	~~12~~

B	***E***	***D***	***C***	***A***

Johnson's Rule Example

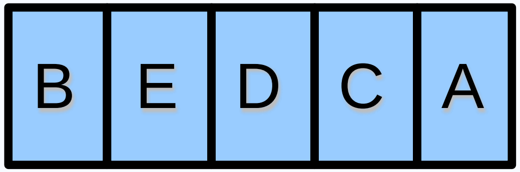
Job	*Work Center 1 (Drill Press)*	*Work Center 2 (Lathe)*
A	5	2
B	3	6
C	8	4
D	10	7
E	7	12

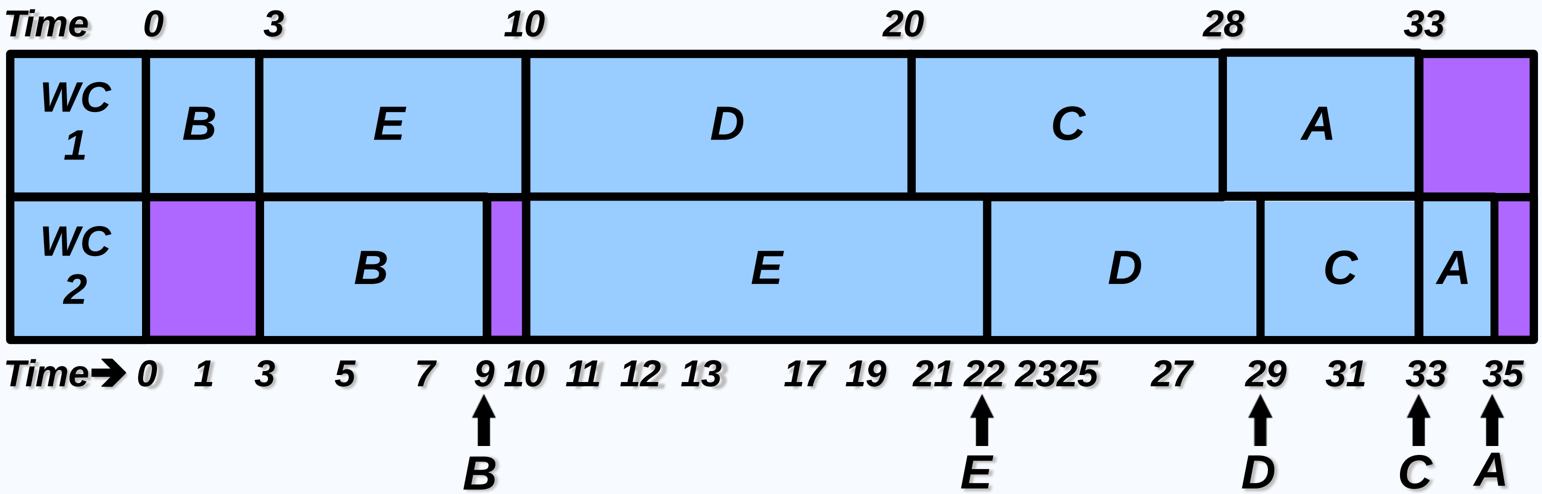




Johnson's Rule Example

Job	Work Center 1 (Drill Press)	Work Center 2 (Lathe)
A	5	2
B	3	6
C	8	4
D	10	7
E	7	12





THANK YOU !

Work Measurement

- **Work study** is the systematic examination of the methods of carrying on activities so as to improve the effective use of resources and to set up standards of performance for the activities being carried out

- **Method study** the systematic recording and critical examination of existing and proposed ways of doing work as a means of developing and applying easier and more effective methods and reducing costs.

- **Work Measurement** is measuring time it should take a qualified worker to complete a specified task, working at a sustainable rate, using given methods, tools and equipment, raw material inputs, and workplace arrangement

Meaningful Standards Help a firm to Determine-

1. Staffing needs(to meet required production)
2. Cost and time estimates prior to production
3. Crew size and work balance(who does what in an assembly line)
4. Expected production(to constitute a fair day’s work)
5. Basis of wage-incentive plans(provides a reasonable incentive)
6. Efficiency of employees and supervision(standard is necessary to determine)

Labor Standards

Four ways of establishing labor standards:

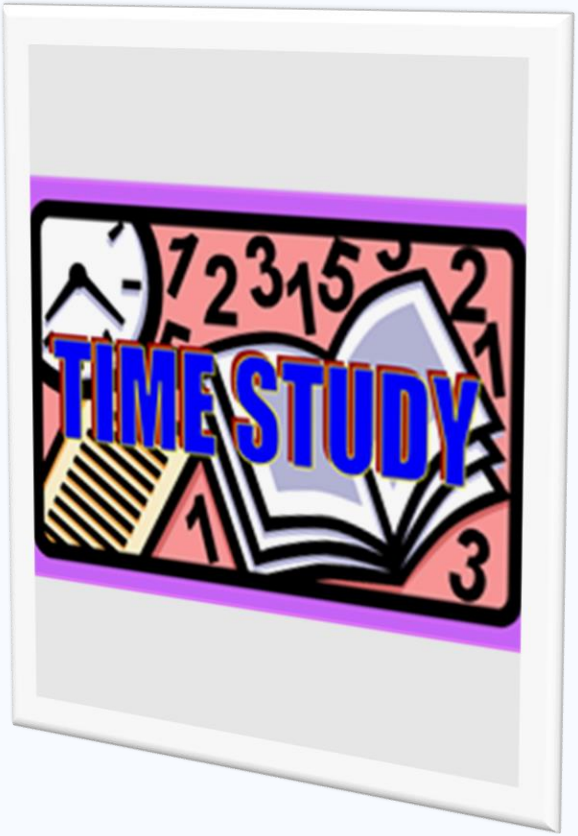
1. Historical experience
2. Time studies
3. Predetermined time standards
4. Work sampling

Historical Experience

- How the task was performed last time
- Easy and inexpensive
- Data available from production records or time cards
- Data is not objective and may be inaccurate
- Not recommended

Time Studies

- Involves timing a sample of a worker's performance and using it to set a standard
- Requires trained and experienced observers
- Cannot be set before the work is performed



Time Studies

1. Define the task to be studied
2. Divide the task into precise elements
3. Decide how many times to measure the task
4. Time and record element times and rating of performance

Time Studies

5. Compute average observed time

$$\text{Average observed time} = \frac{\text{Sum of the times recorded to perform each element}}{\text{Number of observations}}$$

6. Determine performance rating and normal time

$$\text{Normal time} = \text{Average observed time} \times \text{Performance rating factor}$$

Time Studies

7. Add the normal times for each element to develop the total normal time for the task
8. Compute the standard time

$$\text{Standard time} = \frac{\text{Total normal time}}{1 - \text{Allowance factor}}$$

Rest Allowances

- ▶Personal time allowance
  - ▶4% - 7% of total time for use of restroom, water fountain, etc.
- ▶Delay allowance
  - ▶Based upon actual delays that occur
- ▶Fatigue allowance
  - ▶Based on human energy expenditure under various physical and environmental conditions

Time Study Example 1

Average observed time = 4.0 minutes
Worker rating = 85%
Allowance factor = 13%

Standard time??

Time Study Example 1

Average observed time = 4.0 minutes
Worker rating = 85%
Allowance factor = 13%

Normal time = (Average observed time) x (Rating factor)

= (4.0)(.85)

= 3.4 minutes

$$\text{Standard time} = \frac{\text{Normal time}}{1 - \text{Allowance factor}} = \frac{3.4}{1 - .13} = \frac{3.4}{.87}$$

= 3.9 minutes

Time Study Example 2

Allowance factor = 15%

JOB ELEMENT	OBSERVATIONS (MIN)					PERFORMANCE RATING
	1	2	3	4	5	
(A) Compose and type letter	8	10	9	21*	11	120%
(B) Type envelope address	2	3	2	1	3	105%
(C) Stuff, stamp, and seal envelopes	2	1	5*	2	1	110%

Calculate Standard Time

Time Study Example 2

Allowance factor = 15%

	OBSERVATIONS (MIN)					PERFORMANCE RATING
JOB ELEMENT	1	2	3	4	5	
(A) Compose and type letter	8	10	9	21*	11	120%
(B) Type envelope address	2	3	2	1	3	105%
(C) Stuff, stamp, and seal envelopes	2	1	5*	2	1	110%

1. Delete unusual or nonrecurring observations (marked with *)
2. Compute average times for each element

Average time for A = (8 + 10 + 9 + 11)/4 = 9.5 minutes

Average time for B = (2 + 3 + 2 + 1 + 3)/5 = 2.2 minutes

Average time for C = (2 + 1 + 2 + 1)/4 = 1.5 minutes

Time Study Example 2

3. Compute the normal time for each element

Normal time = (Average observed time) x (Rating)

Normal time for A = (9.5)(1.2) = 11.4 minutes
Normal time for B = (2.2)(1.05) = 2.31 minutes
Normal time for C = (1.5)(1.10) = 1.65 minutes

4. Add the normal times to find the total normal time

Total normal time = 11.40 + 2.31 + 1.65
= 15.36 minutes

5. Standard time for the job:

Standard time = Total normal time/(1 - Allowance factor)

= 15.36/(1-.15)

=18.07 min

Thus, 18.07 minutes is the time standard for this job.

Predetermined Time Standards

▶Divide manual work into small basic elements that have established times
▶Can be done in a laboratory away from the actual production operation
▶Can be set before the work is actually performed
▶No performance ratings are necessary

Thank You